

Express Web Servers

Node.js web application, REST API implementation and running-code evidence

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Technology	Node.js · Express · HTML · CSS · JavaScript

Requirement checklist

- ✓ Express server created and running on port 3000
- ✓ Static webpage served from the public folder
- ✓ GET /api/add endpoint adds two values on the server
- ✓ Invalid or missing values return an HTTP 400 JSON error
- ✓ README includes setup, API usage and project structure
- ✓ Automated checks confirm the page and endpoint behaviour

Running application

The browser interface is served by Express from the **public** folder. For this evidence run, the client sent a GET request with 19 and 23. The server returned the result 42.

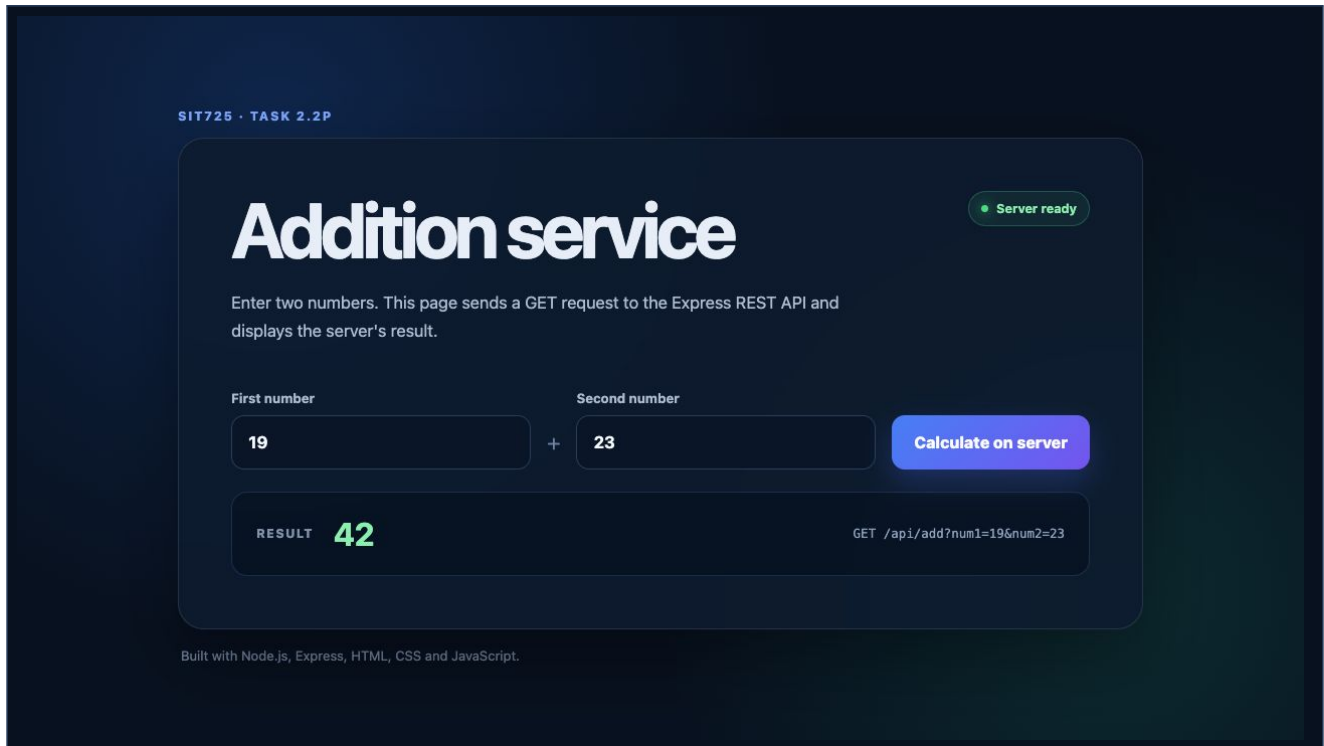


Figure 1. Live Express application showing GET /api/add?num1=19&num2=23 and the server result 42.

How to reproduce

```
npm install
npm start
# Open http://localhost:3000
# API example: http://localhost:3000/api/add?num1=5&num2=7
```

Implementation evidence

The endpoint reads query parameters, validates that both are finite numbers, and returns a JSON response. A clear 400 response is returned when input is missing or invalid.

```
app.get("/api/add", (req, res) => {
  const numbers = parseNumbers(req.query.num1, req.query.num2);

  if (!numbers) {
    return res.status(400).json({
      error: "Please provide two valid numbers using num1 and num2.",
      example: "/api/add?num1=5&num2=7"
    });
  }

  const { num1, num2 } = numbers;
  return res.json({
    num1, num2, operation: "addition", result: num1 + num2
  });
});
```

Automated verification

The project was tested with the built-in Node.js test runner. All three checks passed.

```
$ npm test

✓ serves the public webpage
✓ adds two numbers using the GET endpoint
✓ rejects missing or invalid numbers

tests 3 · pass 3 · fail 0
```

Project structure

```
public/
index.html
script.js
styles.css
test/
server.test.js
.gitignore
package.json
README.md
server.js
```